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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

REGULAR EXAMINATION

March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEFTQ200 - FOOD PACKAGING TECHNOLOGY

(2024 ADMISSION ONWARDS)

Part A

Short Answer type

Answer any 10 questions

Each question carries 2 marks

1. State any two advantages of glass containers for packaging food products.[U]/[CO1]

1. Glass is chemically inert and does not react with food.
2. It provides an excellent barrier against gases and moisture.

2. How does corrosion affect metal cans used for food storage? [A]/[CO1]

Corrosion can weaken the metal surface, cause leakage, and contaminate food with metallic ions, thereby reducing product safety and shelf life.

3. State two uses of paper in food packaging. [U]/[CO1]

1. Wrapping materials for bakery and confectionery products.
2. Production of paper bags and cartons for food storage and transport.

4. Identify one packaging material with high tear strength. Polyethylene (PE) or nylon (polyamide) films. **. [K]/[CO2]**

5. State one difference between MAP and CAP.. [K]/[CO3]

- **MAP (Modified Atmosphere Packaging):** Gas composition is modified before sealing and may change over time.
- **CAP (Controlled Atmosphere Packaging):** Gas composition is continuously monitored and controlled during storage.

6. Explain how biodegradable packaging differs from conventional plastic packaging.. [U]/[CO3]

Biodegradable packaging materials are decomposed by microorganisms into natural substances such as carbon dioxide, water, and biomass, whereas conventional plastics persist in the environment for long periods.

7. What is meant by plastic migration in food packaging?[U]/[CO1]

Plastic migration refers to the transfer of chemical substances such as additives, monomers, or plasticizers from packaging materials into the food product.

8. List one application of stretch wrapping.. [K]/[CO3]

Stretch wrapping is used to secure palletized food products during transportation and storage.

9. Define food packaging.[K]/[CO2,CO5]

Food packaging is the process of enclosing or protecting food products using suitable materials to preserve quality, extend shelf life, and facilitate handling, storage, and distribution.

10. Define intelligent packaging.. [K]/[CO3]

Intelligent packaging refers to packaging systems that monitor the condition of packaged food and provide information about product quality, freshness, or storage conditions.

11. Name any two types of colored glass containers used in food packaging.. [A]/[CO1]

1. Amber glass
2. Green glass

12. Name one method used to evaluate the grease resistance of packaging materials.. [K]/[CO5]

Kit test. [2x10=20]

Part B

Short Essay Questions

Answer any 6 questions

Each question carries 5 marks

13. Explain how retort pouches compare with glass jars in terms of safety and durability..

[U]/[CO1]

Retort pouches are flexible laminated packaging materials designed to withstand thermal sterilization similar to canned foods. They provide excellent barrier properties against oxygen and moisture. Compared with glass jars, retort pouches are lighter in weight, require less storage space, and reduce transportation costs. They also heat more rapidly during sterilization, improving energy efficiency. However, glass jars are more rigid and offer better resistance to puncture and physical damage. Glass is chemically inert and reusable, while retort pouches are usually single use laminated materials. In terms of safety, both packaging systems can effectively preserve food if properly sealed and sterilized.

14. Explain the need for food packaging? [E]/[CO2]

Food packaging plays an essential role in maintaining food quality and safety. It protects food products from physical damage, microbial contamination, moisture loss, and oxidation. Packaging helps extend the shelf life of foods by providing barriers against environmental factors such as oxygen, light, and humidity. It also facilitates storage, transportation, and marketing of food products. In addition, packaging provides important information to consumers, including nutritional content, ingredients, expiry date, and storage instructions. Therefore, effective packaging ensures food safety, convenience, and product integrity.

15. Explain the importance of hermetic sealing in canned food products.[U]/[CO5]

Hermetic sealing refers to airtight sealing of food containers to prevent entry of microorganisms, gases, and contaminants. In canned foods, hermetic sealing is essential to maintain sterility after thermal processing. It prevents recontamination of the product by bacteria, yeast, or molds. A proper hermetic seal also prevents leakage and protects food from

oxygen exposure that may cause spoilage or oxidation. Thus, hermetic sealing plays a crucial role in ensuring safety, stability, and extended shelf life of canned food products.

16. Explain the shelf-life attributes of foods packaged in glass containers.[K]/[CO1]

Glass containers provide excellent barrier properties against gases, moisture, and microorganisms. Since glass is chemically inert, it does not react with food components or alter flavor and aroma. It also prevents oxygen penetration, which helps reduce oxidative spoilage. Additionally, glass protects food from contamination and maintains product integrity. However, exposure to light through clear glass may affect light sensitive foods; therefore, colored glass containers are sometimes used. These properties help maintain the quality and extend the shelf life of packaged foods.

17. Describe the role of retort pouches in extending the shelf life of food products.[U]/[CO5]

Retort pouches are multilayer laminated packaging materials used for thermally processed foods. They allow sterilization of packaged food at high temperatures similar to canning. The multilayer structure provides excellent barriers to oxygen, moisture, and light, which helps prevent microbial growth and oxidation. Retort pouches are lightweight and provide efficient heat transfer during sterilization, ensuring effective destruction of microorganisms. As a result, foods packaged in retort pouches can be stored for long periods at room temperature without spoilage.

18. Write a note on thickness testing of metal sheets used in cans.[U]/[CO2]

Thickness testing is an important quality control measure in the production of metal cans. The thickness of metal sheets affects the strength, durability, and corrosion resistance of the can. Measurement is commonly performed using instruments such as micrometers or thickness gauges. Consistent thickness ensures proper sealing, resistance to pressure during processing, and prevention of leakage or structural failure. Accurate thickness testing helps maintain quality standards in can manufacturing.

19. Define edible films and explain their importance in food packaging..[U]/[CO1]

Edible films are thin layers of edible materials applied to food surfaces to act as protective barriers. They are usually made from proteins, polysaccharides, or lipids. Edible films help

reduce moisture loss, oxygen exposure, and microbial contamination. They also improve food quality, extend shelf life, and reduce the use of synthetic plastic packaging. Since they are biodegradable and consumable, edible films are considered environmentally friendly alternatives in food packaging.

20. Discuss the solubility tests used for identifying common plastics.[U]/[CO5]

Solubility testing is used to identify plastic materials based on their ability to dissolve in specific solvents. Different plastics exhibit characteristic solubility behavior. For example, polystyrene dissolves in aromatic solvents such as benzene or toluene, while polyethylene and polypropylene are generally insoluble in most solvents at room temperature. By observing dissolution or swelling of the plastic sample in various solvents, it is possible to determine the type of plastic used in packaging materials. [5x6=30]

Part C

Long Essay Questions

Answer any 2 questions

Each question carries 10 marks

21. Explain the working principle of aseptic packaging with neat diagrams.[U]/[CO3]

Aseptic packaging is a method in which sterile food products are filled into pre sterilized containers under sterile conditions. The main principle involves separate sterilization of the food product and packaging material before filling.

The process involves the following steps:

1. **Food Sterilization:** The food product is sterilized using ultra high temperature processing to destroy microorganisms.
2. **Packaging Sterilization:** Packaging materials such as cartons or containers are sterilized using hydrogen peroxide or heat.
3. **Aseptic Filling:** The sterile food is transferred into sterilized containers within a sterile chamber.
4. **Sealing:** The container is hermetically sealed to prevent contamination.

This system prevents microbial contamination and allows long term storage of food at room temperature without preservatives. Aseptic packaging is commonly used for milk, fruit juices, and liquid foods.

22. Explain the demerits of plastics in food packaging.[U]/[CO1]

Plastics are widely used in food packaging due to their flexibility and low cost; however, they have several disadvantages.

Environmental Concerns: Plastic waste is non biodegradable and accumulates in the environment, causing pollution and harm to wildlife.

Food Safety Issues: Certain plastic additives such as plasticizers, stabilizers, and monomers may migrate into food products, potentially affecting human health.

Limited Heat Resistance: Many plastics cannot withstand high temperatures, making them unsuitable for certain sterilization processes.

Gas Permeability: Some plastics allow passage of oxygen and moisture, which may reduce shelf life of sensitive foods.

Recycling Challenges: Mixed plastic materials and multilayer packaging are difficult to recycle, contributing to environmental waste problems.

23. Define primary, secondary and tertiary packaging with examples.[K]/[CO1]

Primary Packaging: Primary packaging is the material that directly comes in contact with the food product. Example: Milk bottles, plastic pouches, beverage cans.

Secondary Packaging: Secondary packaging groups multiple primary packages together for protection and handling. Example: Cardboard boxes containing several food packets.

Tertiary Packaging: Tertiary packaging is used for bulk handling, transportation, and distribution of packaged products. Example: Pallets, shrink wrapped cartons used during shipping.

24. Discuss the FSSAI regulations related to food packaging and labelling.[K]/[CO2]

The Food Safety and Standards Authority of India regulates food packaging and labelling under the Food Safety and Standards Act.

Key regulations include:

1. **Safe Packaging Materials:** Packaging materials must be food grade and non toxic.
2. **Prevention of Contamination:** Materials should not transfer harmful substances into food.
3. **Proper Labelling:** Labels must include product name, ingredient list, nutritional information, manufacturing date, expiry date, batch number, and manufacturer details.
4. **Allergen Declaration:** Mandatory labeling of allergenic ingredients.
5. **Storage Instructions:** Clear instructions for safe storage and handling. [10x2=20]